

# Apala Pramanik

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## PERSONAL PROFILE

PhD researcher at UNL specializing in ML-driven beam management and power control for mmWave 5G/6G vehicular networks (V2I/V2X). Experienced in SDR-based hardware-in-the-loop testbeds (USRP/GNU Radio), ray-tracing channel modeling (Wireless InSite, Sionna RT), GPU-accelerated deep learning (CUDA/PyTorch), and Integrated Sensing and Communication (ISAC) systems, with expertise in deep reinforcement learning for wireless PHY-layer optimization.

## EDUCATION

### University of Nebraska–Lincoln – PhD, Computer Science

GPA: 3.75/4.0 | Advisor: Dr. Mehmet Can Vuran, Dale M. Jensen Chair Professor

Dissertation: ML-Driven Beam Management and Power Control for mmWave Vehicular Networks

June 2024 – Present

Nebraska, USA

### University of Nebraska–Lincoln – Master's, Computer Science

GPA: 3.75/4.0 | Advisor: Dr. Dung Hoang Tran, Assistant Professor

Aug 2021 – May 2024

Nebraska, USA

### Guru Gobind Singh Indraprastha University – B.Tech, Electronics and Communication

GPA: 7.9/10 | Thesis: Deep learning techniques for MRI denoising and reconstruction from under sampled MRI scans

Delhi, India

## RESEARCH EXPERIENCE

### University of Nebraska–Lincoln

Graduate Research Assistant – PhD

Nebraska, USA

Aug 2024 – Present

- Camera-Primed Double-Directional Beam Management (IEEE SECON 2026, accepted): Developed a vision-assisted mmWave beamforming framework enabling real-time TX/RX beam selection using camera-derived spatial cues and adaptive correction under hardware constraints.
- Reinforcement Learning for mmWave Power Control: Designed a hardware-in-the-loop RL framework to optimize transmit power and RF gain parameters under SNR constraints using USRP/GNU Radio testbeds and real-time PHY-layer feedback.
- Investigated ISAC principles through bistatic radar characterization at mmWave frequencies, studying dual-function sensing–communication tradeoffs using SDR-based testbeds and controlled scattering experiments.
- Utilized ray-tracing tools (Wireless InSite, Sionna RT) to model mmWave propagation environments and validate experimental channel observations.
- Technical Skills:** GNU Radio, USRP, Python, C++, MATLAB, CUDA/PyTorch, Ubuntu Linux, Git

Graduate Research Assistant – Master's

Aug 2021 – May 2024

- Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems (ACM/IEEE MEMOCODE 2024).
- Applied Neural Network Verification (NNV) for formal analysis of deep learning models in safety-critical systems.
- Specified and verified temporal properties in cyber-physical systems using Signal Temporal Logic (STL).
- Technical Skills:** ROS (rospy, roscpp), Gazebo, RVIZ, Python, C++, Ubuntu Linux, Git

## PUBLICATIONS

### Conference Proceedings

#### Look Once, Beam Twice: Camera-Primed Real-Time Double-Directional mmWave Beam Management for Vehicular Connectivity

Avhishek Biswas, Apala Pramanik, Eylem Ekici, Mehmet Can Vuran

IEEE International Conference on Sensing, Communication, and Networking (SECON), 2026

#### Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems

Apala Pramanik, Sung Woo Choi, Yuntao Li, Luan Viet Nguyen, Kyungki Kim, Hoang-Dung Tran

22nd ACM-IEEE International Symposium on Formal Methods and Models for System Design (MEMOCODE), 2024

#### ASTITVA: Assistive Special Tools and Technologies for Inclusion of Visually Challenged

Apala Pramanik, Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Rohan Tripathi

International Conference on Computing, Communication, and Intelligent Systems (ICCCIS), 2021

#### START: Smart Stick based on TLC Algorithm in IoT Network for Visually Challenged Persons

Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Apala Pramanik

Fourth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud), 2020

### Workshops

#### Vision-based Runtime Monitoring for Human-Construction Robot Systems

Apala Pramanik, Kyungki Kim, Dung Hoang Tran

IROS 2023 Workshop on Formal Methods Techniques in Robotics Systems: Design and Control, 2023

## UNIVERSITY PROJECTS

### mmWave Propagation Engineering and Bistatic Radar Characterization

Spring 2025

- Investigated lateral wave propagation via near-critical incidence on dielectric surfaces; characterized angle-dependent reflection in indoor mmWave environments.
- Built a bistatic radar measurement framework (EGaln/metallic reflectors) to study material-dependent scattering with relevance to ISAC system design.
- Technical Skills:** mmWave Propagation, Bistatic Radar, ISAC, RF Measurement, USRP/SDR, Signal Analysis

### Multi-Modal Traffic Pattern Detection Using Radio and Camera Sensing

Fall 2024

- Built a real-time traffic classification system using synchronized USRP RF signals and camera images; real- and complex-valued CNNs achieved >80% accuracy with <100 ms inference latency via GPU-accelerated execution (CUDA/PyTorch).
- Integrated  $\pm 10$  ms timestamp synchronization and watchdog mechanism for reliable, continuous operation.
- Technical Skills:** Python, PyTorch, CUDA, USRP, Signal Processing, Deep Learning (CNN, CVNN), Multi-Modal Data Fusion

### mmWave 3D Ray Tracing Framework (MATLAB)

Fall 2024

- Built a MATLAB-based 3D ray tracing simulator for mmWave indoor propagation (24 GHz) with multi-bounce reflections, log-normal shadowing, and link budget computation.
- Developed modular environment modeling with power-gradient visualization and automated SVG/EPS/GIF export (Python).
- Technical Skills:** MATLAB, Python, Ray Tracing, mmWave Channel Modeling, Link Budget Analysis

### Deep Learning and NLP Projects

Aug 2021 – July 2024

- Developed a Spanish news classification system using GRU, LSTM, and BETO Transformer models, achieving 91.80% accuracy with tokenizer optimization and data augmentation.
- Implemented CNN image classifiers on Fashion-MNIST and CIFAR-100; sentiment analysis using RNN and Transformer architectures.
- Technical Skills:** Python, PyTorch, TensorFlow, Pandas, Deep Learning (CNNs, RNNs, Transformers)

## INVOLVEMENT

|                         |                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------|
| Peer Reviewer           | IEEE Transactions on Wireless Communications (2025, 2026)                                            |
| Undergrad Research Fair | Presented research to undergraduate Computer Science students at UNL                                 |
| K–12 Outreach           | Demonstrated research activities to K–12 students                                                    |
| Mentorship              | Mentored undergraduate researchers on experimental design, data analysis, and research communication |

## SKILLS

|                |                                                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------------------|
| ML / DL        | PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, CUDA                                                                        |
| Wireless / SDR | GNU Radio, USRP, Wireless InSite, Sionna RT, mmWave, 5G/6G, V2X/V2I, Beamforming, OFDM, Channel Estimation, ISAC              |
| Languages      | Python, C++, MATLAB, Bash/Zsh                                                                                                 |
| Robotics / CPS | ROS (rospy, roscpp), Gazebo, RVIZ, Point Cloud Library                                                                        |
| Tools          | Git, Linux, LaTeX, Arduino, Proteus                                                                                           |
| Coursework     | Deep Learning (A+), Wireless Comm Networks (A+), Advanced Wireless Comm (A), Hardware Acceleration for ML/CUDA (B+), IoT (A+) |
| Soft Skills    | Mentoring, Time Management, Teamwork, Problem-solving, Documentation, Engaging Presentation                                   |

## INTERESTS

|              |                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------|
| Paper Review | Critically evaluating technical rigor, methodology, and scientific contributions in peer-reviewed research. |
| Tech Writing | Conveying complex research clearly and accessibly through papers and documentation.                         |
| Bharatnatyam | Senior diploma holder with 10+ years of training in Indian Classical Dance.                                 |
| Fitness      | Passionate about weight training and maintaining a healthy and active lifestyle.                            |

References available upon request.